

Project #6: Repair HVAC & Controls 446th AW Ops Facility

Project Title: Replace HVAC System in Building J01207

Project Scope:

The existing multizone air handling unit is well beyond its intended life expectancy and is exhibiting repeated mechanical and control failures. Multiple components - including the supply fan assembly, actuators, dampers, and multizone mixing sections - are deteriorated, resulting in inconsistent airflow, poor temperature control, and increased maintenance demand. The unit lacks modern energy-efficient components and does not meet current performance or control standards.

The connected interior finishes - including the carpet, ceiling tiles, and lighting fixtures - are also beyond their service life, with worn flooring, damaged ceiling tiles, and multiple non-functional lights. Because replacement of the HVAC and associated ductwork will require opening the entire ceiling space, it is practical and cost-effective to replace these aging interior elements concurrently rather than performing redundant work in separate projects.

The DDC system requires replacement to interconnect with the HVAC system. The existing DDC system in this building has also experienced multiple failures of the JACE, controllers, and actuators. This system is also obsolete and not compatible with the existing base Energy Monitoring Control System (EMCS) framework version N4 resulting the DDC managers located in Building 2012 inability to control and supervise the building HVAC system activity.

Existing Conditions:

Building J01207 (approx. 11,000 sqft) is currently served by a centralized HVAC system consisting of an 18.6-ton multi-zone packaged air-handling unit that distributes conditioned air to multiple zone throughout the facility. Cooling is provided via a central water chiller, which supplies chilled water to a cooling coil within the packaged unit. Heating is provided by a boiler that supplies hot water to a heating coil located inside the same packaged unit. Zone temperature control is achieved through modulating dampers and control actuators that regulate airflow to individual spaces. See pages 2 and 3 for the existing HVAC system. The system was installed around 2005 and has exceeded its expected service life. Key components – including the supply fan assembly, dampers, and actuators – are experiencing age-related failure, resulting in inconsistent air flow, poor temperature control, and unreliable system operation. The aging system requires increased maintenance effort and does not meet current energy efficiency, contributing to reduced performance and higher operating costs.

Building J01207

Building J01207 is located on the left corner of Barnes Blvd and 12th Street

Legend

- (Reserve) Military and Family Readiness Center
- Hangar 12
- Titus-Will





1207

08 30 2023



1207

446th JMW HEADQUARTERS BUREAU
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08 30 2023



WHA INTERCOMMITTEES ANNEX

08 30 2023



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CONDENSING UNIT						
MARK	MNFR	MODEL NUMBER	TOTAL COOLING MBH	REV. CYCLE HEATING MBH	MINIMUM SEER OR EER BTU/HR. W	NOTES
CU-3	LIEBERT	PFC014A	12.9	-	10.0	1:

NOTES

1. CONDENSING UNIT FOR COMPUTER ROOM AIR HANDLING UNIT.

[illegible]

AIR HANDLING UNITS																																																		
MARK	TYPE		MNFR	MODEL NUMBER	FAN						MOTOR		COOLING COIL										HEATING COIL						RETURN FAN		MAXIMUM OPERATING WEIGHT (LBS)	SOUND POWER DATA RE 10 WATTS OCTIVE BAND CENTER FREQUENCY HZ								NOTES										
	SINGLE ZONE	MULTI ZONE	VAV		BUILT-UP	YORK	TEMPROL	MAX. C.F.M.	MIN. C.F.M.	MIN. O.A. C.F.M.	TYPE	RPM	S.P.		H.P.	R.P.M.	TOTAL MBH	MIN. FACE AREA SQ.FT.	EAT		LAT		ROWS	GPM AT 44° F.W.T.	PRESSURE DROP FT.W.G.	AIR FRICTION IN W.G.	SUCTION TEMP. °F	PRE-HEAT	RE-HEAT	TOTAL MBH		MIN. FACE AREA SQ.FT.	EAT °F	LAT °F	ROWS	LB/HR AT P.S.I. INLET	GPM AT 180° F.W.T.	PRESSURE DROP FT. W.G.	AIR FRICTION IN W.G.		MARK	SEE FAN SCHEDULE								
													TOTAL IN. W.G.	EXTERNAL IN. W.G.					CLASS	DB °F	WB °F	DB °F																					WB °F	DB °F	WB °F					
													63	125					250	500	1000	2000																					4000	8000						
AHU-1	*			*	N/A	4445	4445	600	N/A	N/A	N/A	N/A	II	10.0	N/A	144	11.6	80	67	53.2	52.8	6	26.1	6.78	0.7	-	-	-	251	11.6	54	97	1	-	20.8	3.17	0.1	-	-	N/A	-	-	-	-	-	-	-	1:		
AHU-2	*			*	AP-35	755	530	510	FC	1673	2.44	1.50	II	1.0	1800	32.1	1.51	77.4	63.2	54.9	54.1	6	5.2	1.0	0.39	-	-	-	23.7	2.4	24	55	1	-	2.37	0.7	0.03	-	-	1434	-	-	-	-	-	-	-	-	-	2:
NOTES																																																		
1: EXISTING 5 ZONE AIR HANDLING UNIT, MODEL NUMBER NOT AVAILABLE, HEATING COIL IS EXITING TO REMAIN, CHILLED WATER COIL TO BE PROVIDED UNDER THIS CONTRACT. FIELD MEASURE UNIT TO ENSURE CORRECT COIL IS PROVIDED.																																																		
2: NEW VAV AIR HANDLING UNIT.																																																		

V.A.V. TERMINALS																													
SYMBOL	PRIMARY AIR		TYPE				MFR		MODEL NO.	PRESSURE		DUCT CONN.		AIR HANDLING SYSTEM	HEATING COIL								FAN				CONTROL SEQUENCE	NOTES	
	MAX. CFM	MIN. CFM	SHUTOFF REHEAT	DUAL DUCT	FAN POWERED	CONSTANT VOL.	TRANE	ENVIRO-TEC		TITUS	DESIGN INLET STATIC IN. W.G.	DESIGN DROP IN. W.G. MAX.	INLET		OUTLET	HOT WATER	ELECTRIC	AIR TEMP. ENT/LVG	CAP. MBH OR KW	GPM.	PRESSURE DROP FT. W.G.	WATER TEMP. ENT/LVG	AIR FRICTION-IN. W.G.	PIPE CONNECTIONS	CFM	EXT SP IN WG			HP
VAV-1	455	350	*		*			*	DFVL-2	0.5	0.2	6"ø	12X10	AHU-2	*		68/94	17.3	1.15	0.69	180/150	0.36	1/2	620	0.5	1/6	30		1:2:3
VAV-2	300	180	*		*			*	DFVL-2	0.5	0.2	6"ø	12X8	AHU-2	*		68/93	10.3	0.69	0.50	180/150	0.14	1/2	380	0.3	1/6	30		1:2:3
NOTES																													
1. MAXIMUM AIR PRESSURE DROP FOR TERMINAL IS SUM OF DESIGN DROP AND COIL DROP AT MAXIMUM CFM.																													
2. N.C. LEVEL BASED ON 10db ROOM ABSORPTION, RE RAISED TO THE -12 WATTS.																													
3. PRIMARY AIR VALVE CFM SHALL NOT EXCEED MANUFACTURER'S "NOMINAL CFM".																													

[illegible]

DEPARTMENT OF THE AIR FORCE AIR MOBILITY COMMAND	Designed by: WILLIAMS		Date: JUNE 9, 2003
	Dwn by: BRADSHAW	Chk by: CAMEJO	File Name:
LEGEND AND SCHEDULES	Plot Date:		1" = 1"
	Plot Scale:		
Drawing Number:			

PROJECT NUMBER: PQWY979005	REVITALIZE 446 AW HQ ANNEX - BUILDING 1207
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akewood
ion Parking
Garage

Find address or place

LEGENDS:

COMMUNICATION CABLE

ELECTRIC

WATER

WASTE WATER/SEWER

NOTES:

- NOT TO SCALE

- CONTRACTOR SHALL VERIFY
LOCATION OF UTILITY LINES.

